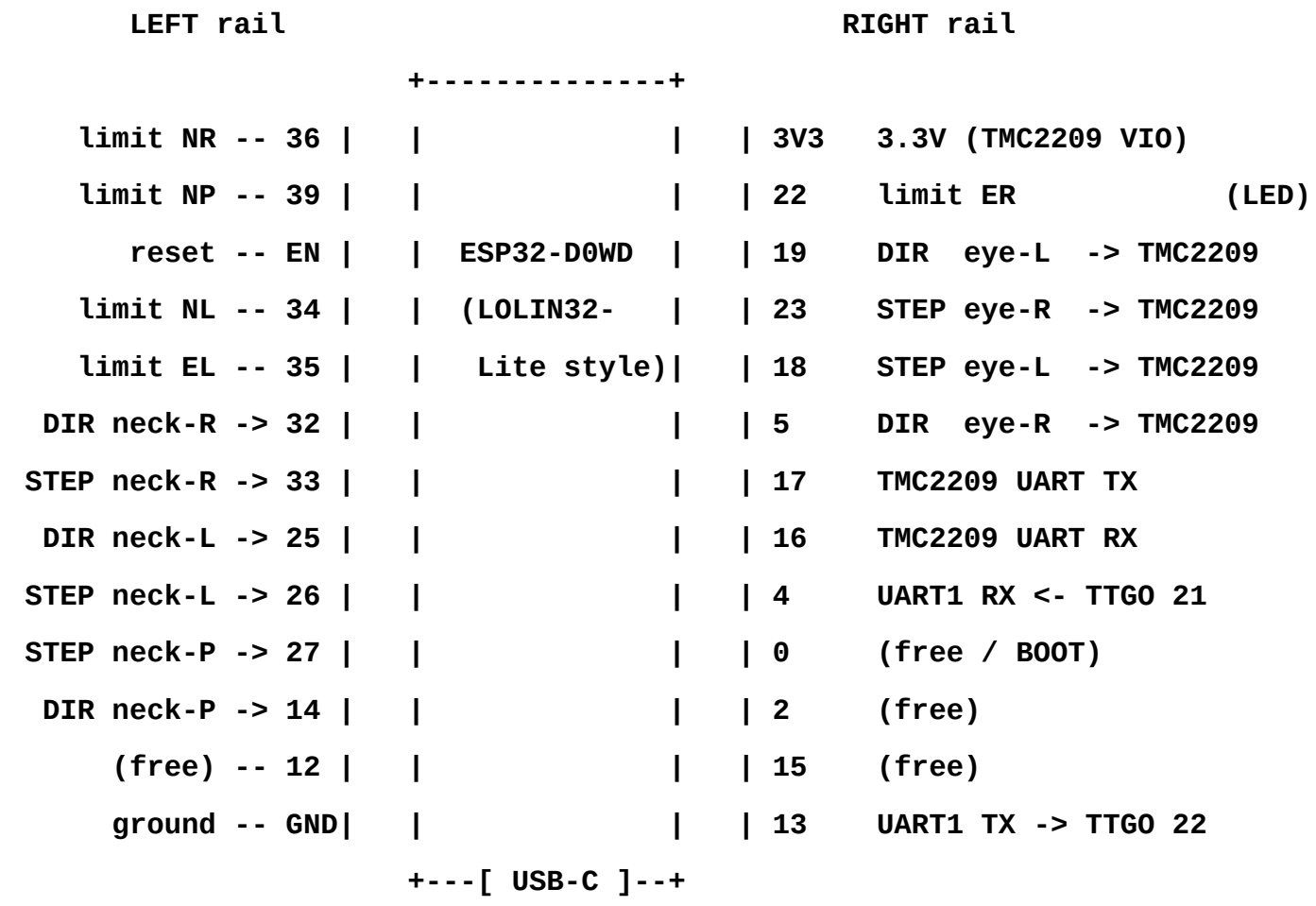
j4_stepper -- pin diagram

ESP32-D0WD (LOLIN32-Lite style), component side up, USB-C at the BOTTOM



Neck STEP/DIR pins feed the DM556TE PUL+/DIR+ inputs (PUL-/DIR- to GND).
Each limit input = two NC switches in series + a 10k pull-up to 3.3V.
GPIO 22's onboard LED doubles as the eye-right in-range indicator.

Pin(s)	Function
26 / 25	neck-left STEP / DIR -> DM556TE PUL+ / DIR+
33 / 32	neck-right STEP / DIR -> DM556TE PUL+ / DIR+
27 / 14	neck-pivot STEP / DIR -> DM556TE PUL+ / DIR+
18 / 19	eye-left STEP / DIR -> TMC2209
23 / 5	eye-right STEP / DIR -> TMC2209
17 / 16	TMC2209 UART TX / RX (shared bus to both eye drivers)
4 / 13	Upstream UART1 RX / TX -- RX from TTGO 21, TX to TTGO 22
34	neck-left limit switches (input, external pull-up)
36	neck-right limit switches (input, external pull-up)
39	neck-pivot limit switches (input, external pull-up)
35	eye-left limit switches (input, external pull-up)
22	eye-right limit switches (input, external pull-up; LED = in-range)